ISW Cross-Correlation Predictions for Energy-Flow Cosmology with DESI DR1 Tracers

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1 Integrated Sachs–Wolfe Predictions

We compute the ISW–galaxy cross-correlation C_ℓ^{Tg} for Energy-Flow Cosmology (EFC) using linear perturbation theory. The matter power spectrum is modelled with the Eisenstein & Hu (1998) no-wiggle transfer function. For the angular projection we adopt the extended Limber approximation with the substitution $k = (\ell + \frac{1}{2})/\chi$. The primary analysis is restricted to $\ell \in [2, 60]$, matching the multipole range used in most ISW analyses; for $\ell \lesssim 10$, where Limber accuracy degrades, we verify that excluding these multipoles does not alter the qualitative conclusions (monotonic $\mathscr{C}$ and amplitude suppression).

The linear growth factor $D(a)$ is obtained by solving the growth equation in $x \equiv \ln a$,

$$D_{xx} + \left(2 + \frac{d\ln H}{dx}\right) D_x - \frac{3}{2}\,\Omega_m(a)\,\mu(a)\,D = 0\,, \tag{1}$$

where EFC enters through a modified Poisson coupling $\nabla^2\Phi = 4\pi G\,\mu(a)\,\bar{\rho}\,\delta$. We parameterise

$$\mu(a) = 1 + \beta\,S(a)\,, \qquad S(a) = \frac{1}{2}\left[1 + \tanh\left(\frac{\ln a - \ln a_t}{\sigma_{\cos}}\right)\right], \tag{2}$$

with $\beta = 0.16$, $a_t = 0.55$ ($z_t \simeq 0.82$), and $\sigma_{\cos} \in [0.20, 0.30]$.

Unless stated otherwise we use a fiducial ΛCDM background $H(a)$; sensitivity tests with a smooth, percent-level background perturbation,

$$w(a) = -1 + \delta_w\,\mathrm{sigmoid}\left(\frac{a - a_t}{\sigma_w}\right), \qquad \delta_w \le 0.05\,, \tag{3}$$

confirm that the inferred $f\sigma_8$ overlap is stable (maximum $\Delta\chi^2 < 1$ on 14 RSD data points).

2 Kernel Decomposition and Cancellation Metric

Using $\Phi(a) \propto \mu(a)D(a)/a$, the ISW kernel can be written as the sum of a standard term and an EFC-specific transition term,

$$K(a) \propto \frac{d}{d\ln a}\left(\mu\,\frac{D}{a}\right) = \underbrace{\mu(a)\,\frac{d(D/a)}{d\ln a}}_{K_\mu} + \underbrace{\frac{d\mu}{d\ln a}\,\frac{D}{a}}_{K_{\mu'}}. \tag{4}$$

By construction $K_{\mu'} \equiv 0$ in ΛCDM. The $K_{\mu'}$ term is non-zero only over a finite interval in $\ln a$ centred on a_t, making its observational impact highly sensitive to the tracer redshift distribution.

We quantify cancellation between the two contributions by defining

$$\mathscr{C} = 1 - \frac{\left|\sum_\ell C_{\ell,\mathrm{tot}}^{Tg}\right|}{\left|\sum_\ell C_{\ell,\mu}^{Tg}\right| + \left|\sum_\ell C_{\ell,\mu'}^{Tg}\right|}\,, \tag{5}$$

with $\mathscr{C} \in [0,1]$. Values $\mathscr{C} \to 1$ indicate near-total cancellation between K_μ and $K_{\mu'}$, which is a distinctive EFC signature for tracers overlapping the transition epoch.

3 Tracer Windows and ISW Amplitude Fitting

We evaluate C_ℓ^{Tg} over $\ell = 2$–60 using DESI DR1–motivated tracer windows of the form $W_g(z) = b_g(z)\, dN/dz$. In this work we approximate dN/dz with truncated parametric forms matched to the published DESI bin edges and effective redshifts; the final pipeline will use the measured DESI redshift histograms. Galaxy biases b_g are adopted from DESI DR1 full-shape analyses (Adame et al. 2025, Table 1).

To compare with observations we fit an ISW amplitude parameter $A_{\rm ISW}$ for each tracer by minimising

$$\chi^2(A_{\rm ISW}) = \sum_\ell \frac{\left(C_{\ell,\rm obs}^{Tg} - A_{\rm ISW}\, C_{\ell,\rm model}^{Tg}\right)^2}{\sigma_\ell^2} \tag{6}$$

over the analysis multipole range. We benchmark against published measurements, including $A_{\rm ISW} = 0.984 \pm 0.349$ from DESI Legacy Survey $\times$ Planck temperature cross-correlation (Hang et al. 2021) and redshift-binned ATLAS LRG $\times$ Planck results (Ansarinejad et al. 2020), including their reported null at $\bar{z} = 0.68$.

4 Key Results

(i) The cancellation index $\mathscr{C}$ increases monotonically with tracer overlap with z_t, from $\mathscr{C} = 0.59$ (LRGz0, $z_{\rm eff} = 0.51$) to $\mathscr{C} = 0.96$ (ELG, $z_{\rm eff} = 1.32$), confirming opposite-sign contributions from K_μ and $K_{\mu'}$ near the transition.

(ii) Bias-calibrated $A_{\rm ISW}$ values predicted by EFC (0.29–0.56) are consistent with all cited measurements within 1.6σ, owing to the large uncertainties in current high-redshift ISW data. The predicted suppression is strongly redshift-dependent, becoming pronounced only for tracers whose redshift distribution overlaps the transition epoch.

(iii) The decisive discriminator is a tomographic C_ℓ^{Tg} measurement in DESI spectroscopic bins overlapping z_t, where EFC predicts large $\mathscr{C}$ (strong cancellation) and suppressed $A_{\rm ISW}$, while ΛCDM has $\mathscr{C} = 0$ by construction.

5 Assumptions and Caveats

- **Limber approximation.** We adopt the extended Limber approximation with $k = (\ell + \frac{1}{2})/\chi$. For $\ell \lesssim 10$, where Limber accuracy degrades, we verify that excluding these multipoles does not alter the qualitative conclusions (monotonic $\mathscr{C}$ and amplitude suppression).

- $A_{\rm ISW}$ **definition.** We report $A_{\rm ISW}$ from a χ^2 amplitude fit, matching the convention used by Hang et al. (2021). The metric is applied to the same ℓ-range for both model and data.

- **Transfer function.** Eisenstein–Hu no-wiggle is used as a proxy; the full pipeline should employ CLASS or CAMB output for both the ΛCDM baseline and EFC.

- **Tracer windows.** Smooth parametric approximations to DESI DR1 $n(z)$; measured histograms should be substituted for the final observational comparison.

- **Galaxy bias.** We use published b_g values rather than fitting from auto-C_ℓ. The implied $W_g(z) = b_g\, dN/dz$ is standard for ISW forecasts.

- **No mask/pseudo-C_ℓ correction.** Observational comparison requires matching the sky mask and estimator used in the data analysis.
- **$\sigma_{8,0}$ treatment.** Fitted as a nuisance parameter per model to isolate the redshift-dependent shape of $f\sigma_8(z)$.

6 Prediction Summary

Table 1 summarises the quantitative EFC predictions alongside ΛCDM expectations.

Observable	**EFC**	**ΛCDM**
Cancellation $\mathscr{C}$	Monotonically increasing toward z_t; $\mathscr{C} \gtrsim 0.6$ for $z_{\rm eff} \gtrsim 0.5$	$\mathscr{C} = 0$ by construction
ISW amplitude $A_{\rm ISW}$	Suppressed near z_t ($A \approx 0.3$–0.6); mild at low z	$A_{\rm ISW} = 1$ for all tracers
Redshift dependence	Strong: suppression peaks at $z_t \simeq 0.82$	Weak: varies only via $\Omega_\Lambda(z)$
ℓ-shape	Only mild ℓ-dependent changes relative to ΛCDM; signature is amplitude, not shape	$S = 1$ (reference)
Key tracer:LRG+ELG ($z_{\rm eff} = 0.93$)	$\mathscr{C} \approx 0.79,\ A_{\rm ISW} \approx 0.29$	$\mathscr{C} = 0,\ A_{\rm ISW} = 1$
Control tracer:LRGz0 ($z_{\rm eff} = 0.51$)	$\mathscr{C} \approx 0.59,\ A_{\rm ISW} \approx 0.56$	$\mathscr{C} = 0,\ A_{\rm ISW} = 1$
Falsification	If tomographic bins overlapping z_t yield $A_{\rm ISW} = 1.0 \pm 0.1$ and $\mathscr{C} \approx 0$, EFC would be ruled out at high statistical significance at the ISW level	If $A_{\rm ISW} \ll 1$ with high $\mathscr{C}$ at $z_t \Rightarrow$ ΛCDM requires modification
Current status	Consistent within 1.6σ	Consistent within $\sim 1\sigma$

Table 1: Summary of EFC and ΛCDM predictions for ISW cross-correlation with DESI DR1 tracers.

Bottom line. EFC and ΛCDM make identical qualitative predictions (ISW signal exists, is positive at low z) but diverge quantitatively for tracers overlapping the transition epoch. The cancellation index $\mathscr{C}$ provides a model-independent diagnostic: any non-zero $\mathscr{C}$ at a given redshift would indicate a time-varying gravitational coupling, and its magnitude directly constrains the transition parameters $(\beta, a_t, \sigma_{\rm cos})$.

A On the Cancellation Index and $F_{\mu'} > 1$

In general, the signed fractional contribution of the $K_{\mu'}$ term,

$$F_{\mu'}^{\rm signed} = \frac{\sum_\ell C_{\ell,\mu'}^{Tg}}{\sum_\ell C_{\ell,\rm tot}^{Tg}}, \tag{7}$$

can exceed unity in absolute value. This is not a pathology but a direct consequence of partial cancellation: when K_μ and $K_{\mu'}$ contribute with opposite signs, the total C_ℓ^{Tg} is smaller than either component individually. In the extreme limit of perfect cancellation ($\mathscr{C} \to 1$), the denominator approaches zero while the numerator remains finite, so $|F_{\mu'}^{\rm signed}| \to \infty$.

For this reason we adopt $\mathscr{C}$ as the primary metric, which is bounded in $[0,1]$ and directly quantifies the degree of cancellation. The signed fraction $F_{\mu'}^{\rm signed}$ is reported as a supplementary diagnostic that reveals the sign structure (both terms contribute with opposite sign for all DESI tracers studied here), but $\mathscr{C}$ should be used for quantitative comparisons.

Physically, the opposite-sign structure arises because near z_t the standard ISW kernel K_μ and the transition kernel $K_{\mu'}$ act in competing directions: the standard term corresponds to the usual late-time decay of gravitational potentials in an accelerating universe, while the transition term modifies this decay through the temporal variation of μ. The net ISW signal is therefore suppressed relative to ΛCDM, and the suppression is strongest for tracers whose redshift window peaks near z_t.

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